

Samba Server

SMB/CIFS File Sharing · Linux System & Network Administration

1. Service Fundamentals

Samba is an open-source implementation of the SMB/CIFS protocol. It allows a Linux server to provide Windows-compatible file and printer shares so that Windows, macOS and Linux clients can map network drives.

Item	Details
Service / Package	Samba · samba
Protocol	SMB / CIFS
Ports	TCP 445 (modern) · TCP 139 (NetBIOS)
Main Config	/etc/samba/smb.conf
Service Units	smbd.service · nmbd.service
Password DB	smbpasswd / tdbsam

2. Architecture

smbd handles file and print sharing plus authentication. nmbd provides NetBIOS name resolution (optional on pure IP networks). Clients authenticate against Samba's user database, which is usually mapped to Linux system users.

3. Installation

Update package list and install Samba:

Command:

```
sudo apt update
```

Command:

```
sudo apt install samba
```

Enable and start the services:

Command:

```
sudo systemctl enable --now smbd nmbd
```

Backup the default configuration:

Command:

```
sudo cp /etc/samba/smb.conf /etc/samba/smb.conf.bak
```

4. Configuration

Open the main configuration file:

Command:

```
sudo nano /etc/samba/smb.conf
```

Example minimal configuration:

```
[global]
workgroup = WORKGROUP
server string = Linux Samba Server
security = user
map to guest = Bad User

[shared]
path = /srv/samba/shared
browseable = yes
read only = no
guest ok = no
valid users = @smbgroup
create mask = 0660
directory mask = 0770
```

Create the share directory and group:

Command:

```
sudo mkdir -p /srv/samba/shared
```

Command:

```
sudo groupadd smbgroup
```

Command:

```
sudo chown root:smbgroup /srv/samba/shared
```

Command:

```
sudo chmod 2770 /srv/samba/shared
```

Add a Linux user to the group and set a Samba password:

Command:

```
sudo usermod -aG smbgroup alice
```

Command:

```
sudo smbpasswd -a alice
```

Command:

```
sudo smbpasswd -e alice
```

Validate configuration and restart:

Command:

```
sudo testparm
```

Command:

```
sudo systemctl restart smbd nmbd
```

5. Verification

Check service status:

Command:

```
sudo systemctl status smbd nmbd
```

Confirm ports are listening:

Command:

```
sudo ss -lntp | grep -E ' :445| :139'
```

List shares from a client:

Command:

```
smbclient -L //SERVER_IP -U alice
```

On Windows map the drive using: \\SERVER_IP\shared

6. Configuration Reference

Directive	Purpose
security = user	Require username and password
path = ...	Directory exported by the share
valid users / write list	Who may access or write
guest ok = yes/no	Allow anonymous access
read only = yes/no	Read-only or read-write
create / directory mask	Default permissions on new files
smbpasswd -a user	Add/set Samba password
testparm	Validate smb.conf syntax

7. Troubleshooting

Problem	Check	Solution
Cannot connect	firewall, smbd, ports	Open TCP 445; restart smbd
Access denied	smbpasswd, valid users, perms	Set Samba password; fix ownership
Share not listed	browseable, workgroup	browseable=yes; match workgroup
Config ignored	testparm · restart	Fix syntax then restart smbd

8. Security

- Use security = user; avoid guest shares for sensitive data.
- Restrict valid users and write list; prefer group-based access.
- Set restrictive create/directory masks.
- Allow TCP 445 only from trusted networks.
- Disable SMBv1 if not required (server min protocol = SMB2).

9. Real-World Scenario

An office needs a shared department folder visible to Windows PCs. The administrator installs Samba, creates /srv/samba/dept, a group, maps users with smbpasswd, opens TCP 445 on the internal firewall, and documents the path \\fileserver\dept.

10. Essential Commands

Command	Purpose
sudo apt install samba	Install Samba
sudo testparm	Validate configuration
sudo systemctl restart smbd nmbd	Apply changes
sudo smbpasswd -a user	Set Samba password
smbclient -L //host -U user	List shares
sudo ss -ltnp grep 445	Check listening port

11. Important Points (Interview / Viva)

1. Samba implements SMB/CIFS so Linux can share files with Windows clients.
2. Main ports: TCP 445 (and historically 139).
3. Config file: /etc/samba/smb.conf · validate with testparm.
4. Users need both a Linux account and a Samba password (smbpasswd).
5. smbd = file/print service; nmbd = NetBIOS name service.